

Unit 3: ARMA/ARIMA Models & Forecasting

Dr Christian Engels
christian_engels@outlook.com

Applied Timeseries in R
7 Feb 2026

Certificate in Engagement with Practice in Finance (CEPF)
Durham University Business School



Autoregressive (AR) Models

Moving Average (MA) Models

ARMA and ARIMA Models

Model Selection

Model Diagnostics

Forecasting

Preview: Volatility Clustering

Exercises

By the end of this unit you will be able to:

- 1 Describe the $AR(p)$, $MA(q)$ and $ARMA(p,q)$ model families
- 2 Use ACF/PACF plots to identify appropriate model orders
- 3 Fit ARIMA models in R with `fable::ARIMA()`
- 4 Evaluate residual diagnostics and compare models via AIC/BIC
- 5 Produce and interpret out-of-sample forecasts

Companion script

Open `Unit-3-ARIMA-Forecasting.R` — it simulates ARMA processes, fits models to real data and produces forecasts.

Autoregressive (AR) Models

An autoregressive model of order p :

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \epsilon_t$$

where $\epsilon_t \sim WN(0, \sigma^2)$.

Key features:

- Current value depends on p lagged values of itself
- Like a regression with lagged y as predictors
- Stationary if all characteristic roots lie outside the unit circle

AR(1) recap (from Unit 2)

$$y_t = c + \phi_1 y_{t-1} + \epsilon_t$$

Stationary when $|\phi_1| < 1$.

Half-life of a shock: $-\ln 2 / \ln |\phi_1|$

Moving Average (MA) Models

A moving average model of order q :

$$y_t = c + \epsilon_t + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q}$$

where $\epsilon_t \sim WN(0, \sigma^2)$.

Key features:

- Depends on current and past *innovations*
- **Always stationary** — finite weighted sum of white noise
- Shocks affect y_t for at most q periods

MA(1) example

$$y_t = \epsilon_t + 0.9 \epsilon_{t-1}$$

ACF: single spike at lag 1, then zero.

PACF: gradual decay.

ARMA and ARIMA Models

The ARMA(p, q) Model

Combining AR and MA components:

$$y_t = c + \underbrace{\phi_1 y_{t-1} + \cdots + \phi_p y_{t-p}}_{\text{AR}} + \epsilon_t + \underbrace{\theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q}}_{\text{MA}}$$

ACF/PACF identification:

	ACF	PACF
AR(p)	Gradual decay	Cuts off at lag p
MA(q)	Cuts off at lag q	Gradual decay
ARMA	Gradual decay	Gradual decay

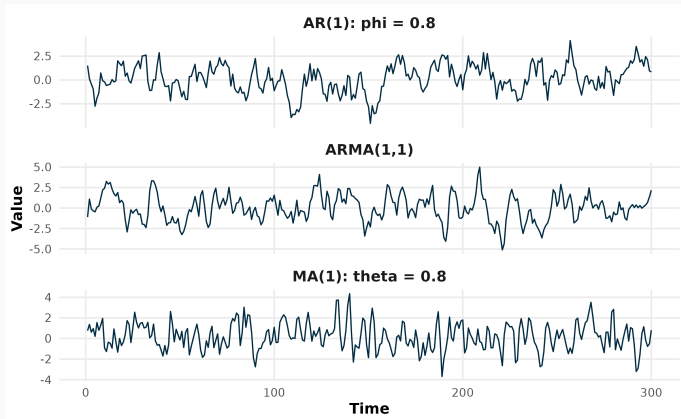
Cuts off: drops to zero. **Gradual decay:** tapers slowly.

ARMA(1,1)

$$y_t = 0.9 y_{t-1} + 0.9 \epsilon_{t-1} + \epsilon_t$$

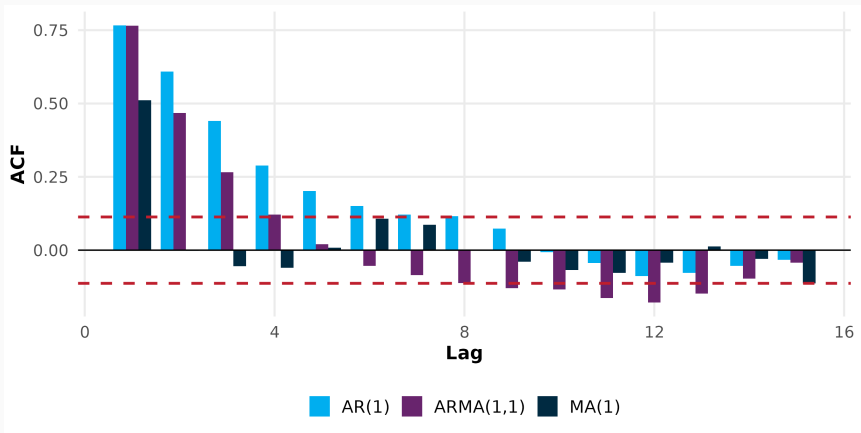
Both ACF and PACF tail off $\Rightarrow$ harder to identify visually.

Simulated ARMA Processes



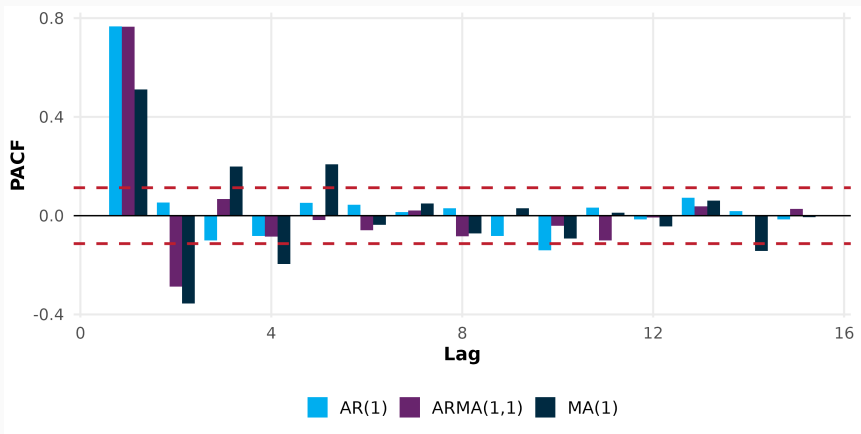
The AR(1) process shows **persistence** — shocks decay slowly. The MA(1) process shows **short memory** — shocks vanish after one period. ARMA(1,1) combines both features.

ACF Fingerprints



AR(1): exponential decay. MA(1): single spike at lag 1, then zero. ARMA(1,1): both tail off. These patterns are the basis for model identification.

PACF Fingerprints



AR(1): single spike at lag 1 (cuts off). MA(1): gradual decay. ARMA(1,1): gradual decay. Compare with the ACF on the previous slide to distinguish AR from MA.

The ARIMA(p, d, q) Model

ARIMA = AutoRegressive Integrated Moving Average.

Apply an ARMA(p, q) model to the d -times differenced series:

$$y'_t = c + \phi_1 y'_{t-1} + \cdots + \phi_p y'_{t-p} + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q} + \epsilon_t$$

where $y'_t = \Delta^d y_t$.

Param.	Meaning
p	AR order
d	Differences for stationarity
q	MA order

Special cases

ARIMA(0,1,0) = random walk

ARIMA($p, 0, 0$) = AR(p)

ARIMA(0,0, q) = MA(q)

Model Selection

Three-step iterative process:

- 1 **Identification:** Use ACF/PACF to propose candidate models
- 2 **Estimation:** Fit candidates and compare AIC/BIC
- 3 **Diagnostics:** Check residuals; if poor, return to step 1

Two approaches

Manual: you identify orders from ACF/PACF, then compare models

Automatic: `ARIMA()` searches for you (convenient but less transparent)

Always inspect the automatic selection with `report()` to verify it makes sense.

To choose between competing models we minimise **information criteria**:

AIC (Akaike):

$$\text{AIC} = -2 \ln \hat{L} + 2k$$

BIC (Bayesian):

$$\text{BIC} = -2 \ln \hat{L} + k \ln T$$

$\hat{L}$: maximised likelihood, k : parameters, T : sample size.

- Lower AIC/BIC $\Rightarrow$ better model
- BIC penalises complexity more heavily
- BIC is **consistent** — selects the true model as $T \rightarrow \infty$

In practice

`glance(model)` returns AIC, AICc, BIC

The `fable` package automates ARIMA model selection:

```
fit <- data %>% model(arima = ARIMA(lreturn))
```

What it does:

- Determines d via unit root tests
- Searches over p and q by stepwise AICc
- Returns the best model automatically

Manual override:

```
ARIMA(lreturn ~ pdq(5,0,1))
```

Useful functions

`report(fit)` — full summary

`coef(fit)` — parameters

`glance(fit)` — AIC, BIC

`accuracy(fit)` — error metrics

AR coefficients (ϕ_i):

- Magnitude $\rightarrow$ strength of persistence
- $\phi_1 = 0.9$: strong dependence on recent past
- $\phi_1 = 0.3$: weak dependence
- Positive = momentum; negative = reversal

MA coefficients (θ_i):

- Controls how past shocks propagate
- $\theta_1 = 0.8$: large shock persistence

Typical financial returns

AR coefficients are usually **small** (close to zero) — reflecting weak predictability of returns.

The model is still useful for capturing residual structure and quantifying uncertainty.

Model Diagnostics

After fitting, verify that residuals resemble **white noise**:

- ① **No autocorrelation:** ACF within bands
- ② **Constant variance:** no patterns in residual plot
- ③ **Normality:** histogram approximately bell-shaped
- ④ **Ljung–Box test:** formal test for residual serial correlation

In R

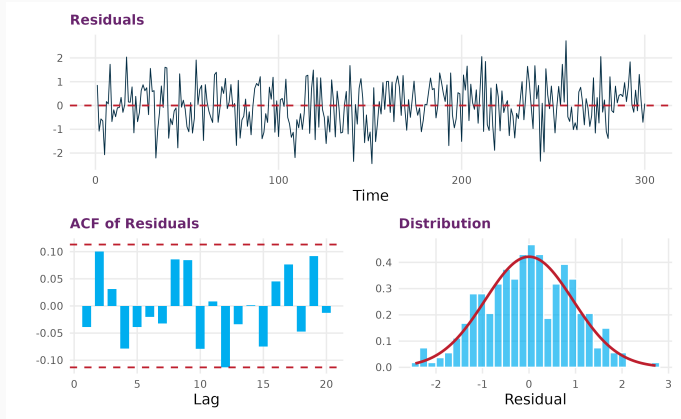
```
fit %>%
```

```
  gg_tsresiduals(lag_max = 12)
```

Patterns in residuals?

Model is mis-specified. Add AR/MA terms or consider GARCH (Unit 4).

What Good Residuals Look Like



Left: residuals fluctuate randomly around zero with no trend. Centre: ACF values fall within the 95% confidence bands. Right: the distribution is approximately normal (red curve).

Forecasting

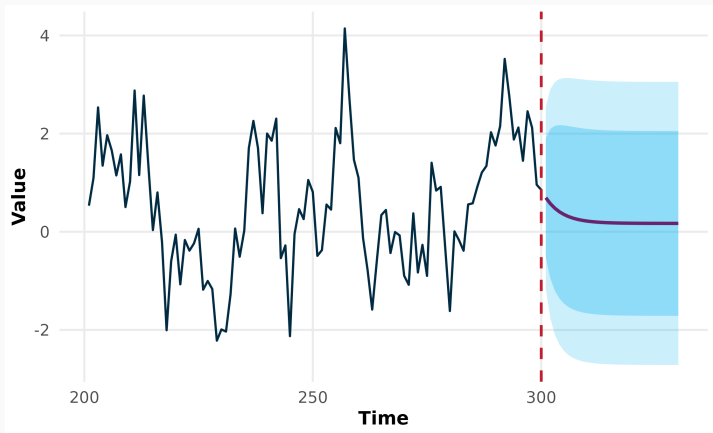
```
fit %>% forecast(h = 12) %>%  
  autoplot(data %>% tail(50))
```

- h: number of periods ahead
- Returns point forecasts with prediction intervals
- Intervals widen with the horizon
- AR forecasts **revert to the mean**

For financial returns

Point forecasts quickly collapse to zero (the unconditional mean). The **prediction intervals** are more informative than the point forecast.

Forecast Fan Chart



The point forecast (purple line) reverts to zero. The prediction intervals (shaded bands) widen as uncertainty grows. For financial returns, these intervals often dominate the point forecast in practical importance.

Preview: Volatility Clustering

ARIMA models capture the **conditional mean** of y_t .

But financial returns often exhibit **volatility clustering**: periods of high variability followed by more high variability.

Squared returns r_t^2 reveal this pattern:

```
data %>%  
  mutate(r2 = lreturn^2) %>%  
  autoplot(.vars = r2)
```

Next up: Unit 4

GARCH models capture the **conditional variance** — modelling how volatility evolves over time.

Exercises

Open `Unit-3-ARIMA-Forecasting.R` — the script provides worked examples before each exercise.

- 1 **Simulation:** (a) Simulate AR(1), MA(1) and ARMA(1,1) processes using `accumulate()`; (b) plot the ACF/PACF for each and verify the identification table
- 2 **Fit:** (a) Download log returns for a stock of your choice; (b) fit an automatic ARIMA with `ARIMA(lreturn)`; (c) use `report()` to inspect the selected order
- 3 **Diagnose:** (a) Run `gg_tsresiduals()`; (b) do the residuals resemble white noise?
- 4 **Forecast:** (a) Forecast 5 days ahead; (b) plot the fan chart — what happens to point forecasts?
- 5 **Preview:** Plot squared log returns. Do you observe volatility clustering?

Hint: Compare `ARIMA(lreturn)` (automatic) vs `ARIMA(lreturn ~ pdq(5,0,1))` (manual) using `glance()`.

You can now:

- Describe AR, MA and ARMA models and their ACF/PACF signatures
- Fit ARIMA models automatically with `fable::ARIMA()`
- Compare models using AIC/BIC and residual diagnostics
- Produce forecasts with prediction intervals
- Recognise that ARIMA captures the conditional mean — volatility requires GARCH (Unit 4)

Next: Unit 4 — Volatility Models (ARCH/GARCH)